Department of Biochemistry

PG PROGRAMME

Choice Based Credit System - Learning Outcomes Based Curriculum Framework (CBCS-LOCF)

(Applicable to the candidate admitted from the academic year 2025 – 2026 onwards)

E. G.S.PILLAY ARTS & SCIENCE COLLEGE (AUTONOMOUS)
NAGAPATTINAM – 611 002. TAMIL NADU
(Accredited by NAAC With "A" Grade)
(MBA – Approved by AICTE, New Delhi)
(Affiliated to Bharathidasan University, Tiruchirappalli – 24)

Department of Biochemistry

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(Applicable to the candidate admitted from the academic year 2025 – 2026 onwards)

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
I	Core Course – I (CC)	Chemistry of Biomolecules	6	5	3	25	75	100
	Core Course – II (CC)	Biochemical Techniques	6	5	3	25	75	100
	Core Course – III (CC)	Cell Biology	6	5	3	25	75	100
	Core Practical – I (CP)	Biomolecules & Biochemical Techniques (P)	6	3	6	50	50	100
	Elective Course – I(EC)	(A) Human physiology (B) Biostatistics	6	4	3	25	75	100
	TOTAL		**30**	**22**				**500**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
II	Core Course – IV(CC)	Enzymology	6	5	3	25	75	100
	Core Course – V (CC)	Molecular Biology	6	5	3	25	75	100
	Core Course – VI(CC)	Metabolism & Regulation	6	5	3	25	75	100
	Core Practical – II (CP)	Enzymology & Molecular Biology(P)	6	3	6	50	50	100
	Elective Course – II (EC)	(A) Bioinformatics (B) Nanotechnology	4	4	3	25	75	100
	Non-Major Elective Course I (NME)	(A) Nutritional Biochemistry (B) Herbal Medicine	2	2	3	25	75	100
	TOTAL		**30**	**24**				**600**

K. UMA, M.Sc., M.Phil.,
Assistant Professor & Head
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Nagapattinam - 611 002.

Dr. C. PRAHALATHAN, M.Sc., Ph.D.
Professor, Department of

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
III	Core Course – VII (CC)	Immunology	6	5	3	25	75	100
	Core Course – VIII (CC)	Research Methodology	6	5	3	25	75	100
	Core Course – IX (CC)	Clinical Biochemistry	6	5	3	25	75	100
	Core Practical – III (CP)	Clinical Biochemistry & Immunology	6	3	6	50	50	100
	Elective Course – III (EC)	(A) Genomics & proteomics (B) Bioethics and IPR	4	4	3	25	75	100
	Internship	Internship Report of the study = 80 marks Viva =20 marks	-	3	-	20	80	100
	Non-Major Elective Course-II (NME)	(A) Detection methods of food adulteration (B) Value addition in food	2	2	3	25	75	100
	TOTAL		**30**	**27**				**700**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
IV	Core Course – X (CC)	Endocrinology	6	5	3	25	75	100
	Core Course-XI (CC)	Pharmaceutical Biochemistry	6	5	3	25	75	100
	Skill Oriented Course-I (SOC)	Medical Laboratory Techniques	6	2	3	25	75	100
	Project	Project work Dissertation = 80 Marks Viva = 20 Marks	12	5	-	20	80	100
	TOTAL		**30**	**17**				**400**
	GRAND TOTAL		**120**	**90**				**2200**

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Professor, Department of
Biochemistry, Bharathidasan

SUMMARY OF CURRICULUM STRUCTURE OF PG PROGRAMME BIOCHEMISTRY

S.No.	Types of the Courses	No. of Courses	No. of Credits	Marks
1	Core courses	11	55	1100
2	Core Practical	03	09	300
3	Elective Courses	03	12	300
4	Non-Major Elective Courses	02	04	200
5	Project Work	01	05	100
6	Internship	01	03	100
7	Skill oriented Course	01	02	100
Total		**22**	**90**	**2200**

EC: I – Semester – I

(a) Human physiology
(b) Biostatistics

EC: II – Semester - II

(a) Bioinformatics
(b)Nanotechnology

EC: II – Semester – III

a) Genomics & Proteomics
b) Bioethics and IPR

NME: I Semester – II

a) Nutritional Biochemistry
b) Herbal Medicine

NME: II Semester – III

a) Detection methods of food adulteration
b) Value addition in food

Programme outcomes

- ✓ PG graduates are professionally competent with characteristic knowledge bank, skill-set, mind-set and pragmatic wisdom in their chosen fields.
- ✓ PG graduates demonstrate the desired sense of being seasoned and exhibit unequivocal spiritedness with excellent qualities of productive contribution to society and nation in the arena of science and technology.
- ✓ PG graduates are mentored such that they exert leadership latitude in their chosen fields with commitment to novelty and distinction.
- ✓ PG graduates are directed in understanding of ethical principles and responsibilities, moral and social values in day-to-day life thereby attaining cultural and civilized personality.
- ✓ PG graduates are able to collate information from different kinds of sources and gain a coherent understanding of the subject.
- ✓

Programme specific outcomes

- ✓ The course aims at gaining an understanding of the processes of metabolic transformation at the molecular level and how these processes are studied.
- ✓ It is important to study enzymes, the rate limiting molecule of all the chemical reactions and understanding enzymes could pave research ideas.
- ✓ Students can make the knowledge of the relationship between structure and function at organ and/or organism level, of important cell biological communication principles and processes, and how they are regulated. Students are able to characterize certain functionalities of biomolecules by using spectroscopic technique
- ✓ Students will gain conceptual understanding of subject matter, scientific reasoning skills, laboratory manipulative skills.
- ✓ The course aims to give participants a basic knowledge of mechanisms of signal transduction and the significance of signal transduction in physiology and pathophysiology.
- ✓ Students can understand the capacity to evaluate and synthesize information from a wide range of sources in order to communicate ideas, concepts and construct arguments in both non-scientific and scientific language.
- ✓ Students will learn at the end of the course to conserve nature and will be able to develop new strategies to preserve the sources of life.

BOS - II

Department of Biochemistry
PG PROGRAMME
Choice Based Credit System - Learning Outcomes Based Curriculum Framework (CBCS-LOCF)
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20/05/2026

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Assistant Professor, Department of Biochemistry,
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Mayiladuthurai – 609 001.

Dr. C. PRAHALATHAN, M.Sc., Ph.D.
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Bharathidasan University Tiruchirappalli – 620 024.

M.Sc., Biochemistry – syllabus- under CBCS-LOCF-candidate admitted from the academic year 2025 - 2026 onwards.

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(a) Bioinformatics
(b)Nanotechnology

EC: II – Semester – III

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b) Bioethics and IPR

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b) Herbal Medicine

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SEMESTER – I

FIRST YEAR	CORE COURSE – I	CREDIT : 5
SEMESTER - I	CHEMISTRY OF BIOMOLECULES	P25BCCC11

COURSE OUTCOMES :

The main objectives of this course are to:

- Students will be introduced to the structure of biomolecules.
- The significance of carbohydrates in biological processes will be understood.
- The structure, properties and biological significance of lipids in the biological system will be studied
- Students will learn about the concepts of protein structure and their significance in biological processes and creatively comprehend the role of membrane components with their biological significance.
- Students will gain knowledge about the structures and functional roles of nucleic acids in the biological system.

COURSE SPECIFIC OUTCOMES:

- The students will be able to understand the source, chemical structure, properties, function and uses of various polysaccharides.
- The students will be able to understand amino acid structures, their physical and chemical properties, and primary, secondary, tertiary and quaternary structure of proteins..
- The students will understand the structure of nucleic acids and its types
- The students will understand the biological importance of vitamins and minerals in the biological system.

UNIT - I **Hours 18**

HOMO AND HETEROGLYCANS: Polysaccharides occurrence, structure, properties and functions of homoglycans starch, glycogen, cellulose, dextrin, inulin, and chitin. Occurrence, structure, properties and functions of heteroglycans, hyaluronic acid, keratan sulphate and chondroitin sulphate. Bacterial cell wall polysaccharides, Blood group substances, Sialic acid, Glycoproteins, Lectins biological functions.

UNIT - II **Hours 18**

PROTEINS: Classification, structure and properties of amino acids, Essential and non-essential amino acids. Non protein amino acids. Proteins Classification based on solubility, shape, composition and function. Properties of proteins. Denaturation and renaturation of proteins. Structure of peptide bonds. Chemical synthesis of polypeptides. Protein structure Primary, secondary, tertiary and quaternary structures of protein. Forces stabilizing the secondary, tertiary and quaternary structures of proteins. Ramachandran Plot Structure and biological functions of fibrous proteins (keratins,), globular proteins (haemoglobin, myoglobin).

UNIT - III **Hours 18**

LIPIDS:Definition and classification of lipids. Fatty acids classification, nomenclature, structure and properties. Triacylglycerols. Classification, structure and function of prostaglandins. Chemical properties and functions of phospholipids and their structures lecithins, cephalins, phosphatidylserine, phosphatidylinositol, plasmalogens. Glycolipids (cerebrosides and gangliosides), Isoprenoids and sterols (cholesterol), bile acids and bile salts. Types and functions of plasma lipoproteins

UNIT - IV **Hours 18**

NUCLEIC ACIDS: Structure of purines and pyrimidines. Components of nucleic acids nucleosides, nucleotides, and polynucleotides. Properties of DNA: buoyant density, viscosity, hypochromicity, denaturation and renaturation- the cot curve. Watson-Crick model-Primary, secondary and tertiary structures of DNA. Triple helix and quadruplex DNA. Mitochondrial and chloroplast DNA. Determination of nucleic acid sequences by Maxam Gilbert and Sanger's methods. Forces stabilizing nucleic acid structure. structure of different types OF DNA(A, B and Z DNA), biological role. Structure and role of different types of RNA.

UNIT - V **Hours 18**

VITAMINS & MINERALS – Definition, sources, functions, deficiency syndromes of Fat soluble vitamins (A, D,E and K) and Water soluble vitamins (B complex and C).and its Structure and biochemical properties . Macro minerals (Ca, P, Mg, Na, K, Cl) and microminerals/trace elements (Co, I, Fe, Mn, Zn, and F) - their sources, daily requirements, functions and deficiency diseases symptoms..

Reference Book

1. L. Stryer. 2002. Biochemistry, 5th edition. W.H.Freeman Company, New York 2. Horton, R., Moranm, LA., Scrimgeour, G, MarcPerry and David Rawn. 2006. Principles of Biochemistry, 4thedition.
2. L. Lehninger, Nelson &Cox. 2007. Principles of Biochemistry, 5thedition, CBS, India.
3. Murray R. K. et al. Harpers Illustrated Biochemistry, 2009, 28thedition. Lange Medical Books/McGraw-Hill.
4. Zubay, 2005. Principles of Biochemistry, 4thedition. Prentice hall.
5. Richard Harvey, Denise Ferrier. 2005. Lippincott. Outlines of Biochemistry, 5thedition.

BOS CHAIRMAN
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FIRST YEAR	CORE COURSE - II	CREDIT : 5
SEMESTER - I	BIOCHEMICAL TECHNIQUES	P25BCCC12

COURSE OUTCOMES:

Biochemical techniques combine various inter-disciplinary methods in biological research and the course aims to provide students with the following objectives:

- To understand the various techniques used in biochemical investigation and microscopy.
- To explain chromatographic techniques.\ and their applications
- To explain electrophoretic techniques.
- To comprehend the spectroscopic techniques and demonstrate their applications in biochemical investigations.
- To acquire knowledge of radio labelling techniques and centrifugation.

COURSE SPECIFIC OUTCOMES

- To be able to communicate and discuss General principles of biochemical investigation Familiarity with working principles, tools and techniques of analytical techniques
- Students can understand the knowledge for the separation of Proteins/peptides by selecting appropriate separation techniques.
- Students are able to characterize certain functionalities of biomolecules by Using spectroscopic techniques.

UNIT - I

Hours 18

UNITS OF MEASUREMENT OF SOLUTES IN SOLUTION: Normality, Molarity, molality and milliosmol. Ionic strength. pH, POH, Henderson Hesselbalch equation, buffers, pH of body fluids. Measurement of pH by indicators, Zwitterions .pH dependent ionization of amino acids and proteins. Viscosity, surface tension and Donnan membrane equilibrium. Principles of electrochemical techniques measurement of pH by glass electrode and hydrogen electrode. Oxygen electrode principles, operation of a Clarke electrode and its applications

.UNIT - II

Hours 18

CELL FRACTIONATION TECHNIQUES AND RADIO ISOTOPE TECHNIQUES: Cell lysis, differential and density gradient centrifugation, Salting in, Salting out, Dialysis, Ultrafiltration. Ultra Centrifugation preparative and analytical ultracentrifuge, Svedberg's constant, Sedimentation velocity and Sedimentation equilibrium, Schlieren optics. Radioisotope technique: Radioactive decay constant, half life of an isotope, Detection and measurement of radioactivity, Geiger Muller counters, scintillation counting, auto radiography and RIA, Application of isotopes in biological studies..

UNIT - III

Hours 18

CHROMATOGRAPHIC TECHNIQUES: Principles, Instrumentation and Applications of Paper, Column, Thin Layer Chromatography, Adsorption, Ion exchanges, Gel filtration, Affinity,

Gas Liquid Chromatography ,Chromato focusing, High Pressure Liquid Chromatography, Fast Protein Liqud Chromatography.

UNIT - IV **Hours 18**

ELECTROPHORETIC TECHNIQUES: Polyacrylamide gel electrophoresis, SDS-PAGE, 2D PAGE, Isoelectric focusing, Visualizing protein bands CBB & Silver staining. Agarose gel Electrophoresis, pulse field electrophoresis, high voltage electrophoresis, Capillary Electrophoresis, Restriction Fragment Length Polymorphism, Florescence In Situ Hybridization. Blotting techniques -Western, Northern & Southern
and its applications.

UNIT - V **Hours 18**

SPECTROSCOPIC TECHNIQUES: Spectroscopic technique: Colorimetry, spectrophotometry UV & visible, Principle Beer Lambert's law, Extinction coefficient. Principle Instrumentation and application Atomic Absorption Spectroscopy, Fluorimetry. Basic principle and application of mass spectra, Nuclear Magnetic Resonance, Electron Spin Resonance, Mass Spectrometry, Matrix Assisted Laser Desorption/Ionization-Time Of Flight, Magnetic Resonance Imaging, Computed Tomography scans. Biochips (DNA chips, Protein chips and Sensor chips). Vibration Spectra IR Applications. X-ray crystallography protein crystals,

Reference Book:

1. Wilson and Walker. A biologists guide to principles and techniques of practical biochemistry. 5th ed.Cambridge University Press 2000.
2. Boyer, R. Modern Experimental Biochemistry. 3rd ed. Addison Weslery Longman, 2000.
3. Upadhyay, Upadhyay and Nath. Biophysical Chemistry Principles and Techniques. Himalaya Publ. 1997.
4. Simpson CFA & Whittaker, M. Electrophoretic techniques.
5. Sambrook. Molecular Cloning. Cold Spring Harbor Laboratory, 2001.
6. Friefelder and Friefelder. Physical Biochemistry Applications to Biochemistry And Molecular Biology. WH Freeman &Co. 1994. 7. Pavia et al. Introduction to Spectroscopy. 3rd ed. Brooks/Cole Pub Co., 2000.

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FIRST YEAR	CORE COURSE – III	CREDIT : 5
SEMESTER - I	CELL BIOLOGY	P25BCCC13

COURSE OUTCOME:

- The Cell Biology course will enable you to consider the most exciting current problems in the field. This course will give you a unique opportunity to study the mechanisms that define and regulate the function of cells and organisms.
- As this is a research-focused master's course, one will take an interactive approach to learning, rather than taking traditional lectures, through seminars, small group tutorials and research placements.

COURSE SPECIFIC OUTCOME:

- Students will demonstrate their mastery of cell biology concepts.
- Learn the most significant functions of cell organelles in molecular level.
- Acquire knowledge on internal organization of cell.
- Understand the concepts of cell cycle, Regulation of cell
- Understand the concepts of cell cycle, Regulation of CDK check points. Apoptosis and Cancer.

UNIT – I **Hours 18**

THE CELL: Cell theory, Protoplasm and Organismal theory, Broad classification of cell types –Bacteria, Archaea (Prokaryotes) and Eukaryotes. Cell Membranes: Basic properties of Cell Membranes – The lipid Bilayer: Composition and Structural Organization, Membrane proteins: Structure and basic functions, Membrane transport of small molecules and the electrical properties of Membranes, Endocytosis and Exocytosis. Specialized structures - Cell Junctions – Occluding, Anchoring and Gap, Ion channels. The plant Cell wall.

UNIT – II **Hours 18**

CELL ORGANELLES; Organelles bounded by Double Membrane Envelopes [Nucleus and Mitochondria], Organelles bound by single membrane [Peroxisome, ER, GA, Lysosomes], Ribosomes, Dictyosomes, Microbodies, Peroxisome, Plastids- Chloroplast, Chromoplast and Leucoplast. Vacuoles and Centrosomes.

UNIT – III **Hours 18**

INTERNAL ORGANIZATION OF THE CELL:The Cytoskeleton: Components of Cytoskeleton, Structure and basic functions of Microtubules, Microfilaments, Intermediate filaments. Cell Communication: Cell-Cell Junctions, Tight Junction, Gap Junction, Cell – Matrix Anchoring junctions: Desmosomes, Adhering Junctions. Cell-Cell Adhesion proteins: Cadherins, Catenins, Integrins, Selectins. Plasmodesmata in Plants. The extracellular Matrix of Animals, Collagens, Elastins, Fibronectin, Laminin.

UNIT – IV **Hours 18**

CELL CYCLE; Overview of Cell cycle: Mitosis and Meiosis, Stages of Mitosis and Meiosis, Crossing over and Linkage. Model organisms and methods to study the cell cycle. Regulation of CDK activity and role of checkpoints. Apoptosis and Cancer.

UNIT – V **Hours 18**

STEM CELLS; Overview/Concepts: Introduction to Concepts in Stem Cell Biology-Potency,Plasticity, Self-Renewal and Expansion, Properties, Stem Cells. Classification andSources. Embryonic Stem cells, Stem cell Morphology Differentiation, Origin and Types of Stem cells. Stem cells and Tissue Renewal.

TEXT BOOK

1. Stephen R. Bolsover, Elizabeth A. Shephard, Hugh A. White, Jeremy S. Hyams. Cell
2. Biology – A short Course. Third Edition, Wiley-Blackwell Publication, 2011
3. George Plopper, David Sharp, Eric Sikorski. Lewin's Cells – THIRD Edition, Johnes&Bartlett Student Edition, 2015.
4. Lodish, Harvey et al., ‚Molecular Cell Biology', 7th Edition, W.H.Freeman, 2005
5. Cooper, G.M. and R.E. Hansman ‚The Cell: A Molecular Approach', VIIth Edition, ASM Press, 2007
6. Sadava, D.E. ‚Cell Biology: Organelle Structure and Function', Panima Publishing, 2004
7. Stem Cells- Scientific Facts and Fiction (Elsevier) By Christine Mummery, Sir Ian Wilmut,
8. Anja van de Stople, Bernard A J Roelen. 2011.

Recommended References:

1. Alberts, Bruce et al., ‚Essential Cell Biology', IVth Edition, Garland Press (Taylor &Francis), 2004.
2. Bruce Alberts et al., Molecular Biology of the Cell. Sixth Edition, 2015
3. Becker, W.M. et al., ‚The World of the Cell', 9th Edition, Pearson Education,

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FIRST YEAR	CORE PRACTICAL - I	CREDIT : 3
SEMESTER - I	BIOMOLECULES & BIOCHEMICAL TECHNIQUES (P)	P25BCCP1

COURSE OUTCOMES :

- To assay the activity of enzymes from different sources.

COURSE SPECIFIC OUTCOMES

- To stimulate their interest in learning the structure, function and kinetics of enzymes and their role as catalyst and regulator of cell metabolism and to serve as foundation for more advanced enzymology courses.

EXPERIMENTS: HOURS : 90

1. Estimation of glucose by Benedict 's method
2. Estimation of proteins by Lowry / Bradford method
3. Estimation of phospholipids by phosphorus assay
4. Estimation of Ascorbic Acid
5. Separation of Subcellular Fractionation By Gradient Centrifugation (Nucleus, Mitochondria)
6. Estimation of DNA by diphenyl amine method
7. Estimation of RNA by orcinol method.
8. Desalting of proteins by dialysis
9. Separation of polar and non polar lipids by TLC
10. Rf value calculation of various amino acids using TLC and PC
11. Separation of proteins by SDS PAGE

REFERENCES:

1. Laboratory manual for Analytical Biochemistry & Separation Techniques, P.Palanivelu, MKU University, Madurai. 2001. 2. Introductory practical Biochemistry S.K. Sawhney, Randhir Singh, 2nd ed,2005.
2. Biochemical methods - S.Sadasivam, New Age International Pub, 2000.
3. Instrumental Methods of Chemical Analysis Bk.Sharma, Goel publications,Meerut, 2000.
4. Enzyme Kinetics A modern Approach. AG Marangani, John Wiley & Sons, 2003.
5. Laboratory Manual in Biochemistry, Jayaraman, New Age International Pub, 2000.

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SECOND YEAR	ELECTIVE COURSE – I A	CREDIT : 4
SEMESTER - I	HUMAN PHYSIOLOGY	P25BCE1A

COURSE OUTCOMES:

- Physiology may be defined as the study of the functions of living organisms.
- It address on how organs and systems within the human body perform their functions at the molecular and cellular level, and the impact of these functions on the entire human body.
- This course is designed to provide students with an understanding of the function & regulation of the human body and physiological integration of the organ systems to maintain homeostasis.

COURSE SPECIFIC OUTCOMES

- At the end of the course students should.
- Have an enhanced knowledge and appreciation of mammalian physiology.
- Understand the functions of important physiological systems including the cardio-respiratory, renal, reproductive and metabolic systems.
- Understand how these separate systems interact to yield integrated physiological responses to challenges such as exercise, fasting and ascent to high altitude, and how they can sometimes fail.
- Be able to perform, analyze and report on experiments and observations in physiology

UNIT - I **Hours 18**

BODY FLUIDS AND CIRCULATORY SYSTEM 8 Lectures Characteristics of Life, Tissues/Organ Systems. Extracellular fluid, Intracellular fluid: Lymph & Blood. Osmolarity of the body fluids, ionic composition, electrolytes, body buffers. Circulation: Structure and functions of Heart and blood vessels, cardiac cycles, cardiac factors controlling blood pressure, artherosclerosis, myocardial in fraction electrocardiogram. Respiration: Anatomy and physiology of respiration, pulmonary surfactant, Gas Exchange & Transport: exchange of gases between lung and blood and between blood and tissues. Role of lung in acid-base balance

UNIT - II **Hours 18**

SKELETAL AND MUSCULAR SYSTEM 8 Lectures Physiology of bone, ligaments and tendons. Role of minerals in bone strength. Osteoporosis, rickets. Types and functions of muscles and muscle proteins. Mechanism of muscle contraction. Retinal physiology: Anatomy and

Function of Retina, Rhodopsin - Retinal Visual Cycle. The Sense of Hearing, Tympanic Membrane and the Acicular System, Chemical Senses -Taste and Smell

Hours 18

UNIT - III

NERVOUS SYSTEM 8 Lectures Physiology: Nervous Tissue & Organization. Structure of neurons, axon and dentrites. Sensory Physiology; Peripheral Nervous System, Central nervous system. Membrane Potentials, Impulse Communication and Nerve Impulse Conduction; Resting potential and action potential, Synaptic transmission. Brain - chemical composition, metabolism, metabolic adaptation, neurotransmitters and cAMP. Biochemical aspects of learning and memory. Enkephalins and endorphins.

Hours 18

UNIT - IV

DIGESTIVE SYSTEM 8 Lectures Digestive system: Anatomy of the digestive system, Salivary, Gastric and Biliary Secretions- composition and functions. Pancreatic secretions. Ingestion of food, Gastrointestinal hormones, movements in Gastro intestinal tract, Digestion and absorption of carbohydrates, lipids and proteins in small intestine and large intestine.

Hours 18

UNIT - V

EXCRETORY SYSTEM 8 Lectures Excretory system: Structure and functions of kidney. Urine - composition and formation. Renal regulation of acid-base balance. Role of kidney in excretion. Determinants of the GFR, Renal Blood Flow, Physiologic Control of Glomerular Filtration and Renal Blood Flow. Endocrinology: Basics of hormones, Endocrine glands, hormones produced by endocrine glands. Physiology of hormones. Mechanism of hormonal action. Hormonal disorders.

TEXTBOOKS

1. Widmaier, Raff, & Strang: Vander's Human Physiology: The Mechanisms of Body Function, 12th edition, McGraw Hill, 2008. ISBN 978-0-07-337810-7

2. McArdle, Katch & Katch, edition, Exercise Physiology: Nutrition, Energy, and Human Performance, seven edition, Lippincott Williams & Wilkins, 2010. ISBN 978-0-7817-9781-8

3. Guyton and Hall, Medical Physiology, 11th edition, Elsevier-Saunders, 2006. problems,

4. Textbook in Medical Physiology and Pathophysiology: Essentials and clinical Copenhagen Medical Publishers 1999 – 2000 Access at: http://www.mfi.ku.dk/PPaulev/content.htm.

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FIRST YEAR	ELECTIVE COURSE – I B	CREDIT : 4
SEMESTER - I	BIOSTATISTICS	P25BCE1B

COURSE OUTCOMES:

- The course emphasizes various statistical methods and its significance.
- The students are expected to understand the concepts and solve relevant problems pertaining to each topic.
- To provide sufficient background to be able to interpret statistical results in research.

COURSE SPECIFIC OUTCOMES:

- Students will able to describe various application area of biostatistics
- Will able to distinguish different types of data and sampling techniques
- To calculate and interpret measures of central tendency and variability in Statistical data
- Will able to compute and interpret the result of correlation and regression analysis
- To compare different population sample using ANOVA.

UNIT - I **Hours 18**

STATISTICAL SURVEY: Organizing, planning and executing the survey. Source of data Primary and secondary data, collection, observation, interview, enquiry forms, questionnaire schedule and checklist. Classification and tabulation of data. Diagrammatic and graphical presentation of data.

UNIT – II. **Hours 18**

MEASURES OF CENTRAL TENDENCY: Arithmetic mean, median, mode, quartiles, deciles and percentiles. Measures of variation range, quartile deviation, mean deviation, standard deviation, Coefficient of variation. Correlation analysis Scatter diagram, Karl's Pearson's coefficient of correlation and Spearman's rank method. Regression analysis.

UNIT III **Hours 18**

PROBABILITY: Definition, concepts, theorems (proof of the theorems not necessary) and calculations of probability Simple problems. Theoretical distributions -Binomial, Poisson and normal distribution Simple problems (proof of the theorems not necessary).

UNIT - IV **Hours 18**

SAMPLING DISTRIBUTION AND TEST OF SIGNIFICANCE:Concepts of sampling, Testing of hypothesis, errors in hypothesis testing, standard error and sampling distribution, sampling of variables (large samples and small samples.). Student's "t"distribution and its applications. Chi-square test and goodness of fit. Analysis of variance one way and two way classification. Duncan's Multiple Range test. Student's t test, ANOVA: Comparison of means in one or two groups (student's t-test). Comparison of means in three or more groups (ANOVA), F- test

UNIT - V **Hours 18**

DATA REPRESENTATION & ANALYSIS:Design of experiment- Completely randomized block design, Randomized block design. Histogram, Stem-&-Leaf Plot, Line Diagram, Frequency Polygon, Frequency Curve, Pie Diagram, Bar Diagrams, Scatter Diagram, Box-&-Whisker Plot, Bubble Plot, Growth chart, Dendrogram, Nomogram, Partogram, Pedigree Chart, Cartogram.

Reference Book:

1. Statistical Methods, 4th Edition- Gupta, S.P, Sultan Chand &Son Publishers. 2012.
2. Biostatistical Analysis, 5th Edition- Zar, J.H, Pearson Education, 2010. 3. Biostatistics Daniel, W.W. A Foundation for Analysis in Health Sciences, 10th Edition, John Wiley and Sons, Inc., 1999.

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SEMESTER – II

FIRST YEAR	CORE COURSE - IV	CREDIT : 5
SEMESTER - II	ENZYMOLOGY	P25BCCC21

COURSE OUTCOMES :

- Students will obtain basic knowledge about the relationship between properties and structure of the enzymes, their mechanism of action and kinetics of enzymatic reactions.
- Students can understand to compare and contrast the historical uses of enzyme technology with current applications in a diverse range of industries

COURSE SPECIFIC OUTCOMES

- The student was able to analyze the structure/function relationships in biocatalyzed reactions.
- Students are able to research a contemporary application of enzyme technology or metabolic engineering and present the results in a well-structured oral presentation.
- At the end of the course students will be explored to understand the use of enzymes in medicine, food, organic synthesis, genetics and other areas that favor a wide reach for them.

UNIT - I **Hours 18**

INTRODUCTION OF ENZYMES: Definitions (catalytic power, specificity, reactivity, regulation, Holoenzyme, Apoenzyme, coenzymes and cofactors). Enzyme Nomenclature and IUB system of enzyme classification. Active site- Fisher and Koshland models. Formation of enzyme substrate complex evidence. Basic concepts of bioenergetics: The collision theory, activation energy and transition state theory. Measurement and expression of enzyme activity enzyme assays, Enzyme units.

.UNIT – II **Hours 18**

ENZYME KINETICS: Pre-Steady state and Steady state kinetics,. Kinetics of single substrate enzyme catalyzed reaction Michaelis-Menten (Briggs- Haldane) equation, Double-Reciprocal Plot, Lineweaver Burk plot, and Eadie-Hofstee plot. Determination of Vmax, Km, Kcat, Specificity constant (Kcat/Km) and their significance. Factors influencing enzymatic activity. Kinetics of enzyme reactions having two or more substrates. Single displacement and double displacement

UNIT - III **Hours 18**

MECHANISM OF ENZYME ACTION AND ENZYME INHIBITION: Mechanism of enzyme action: Factors contributing to the catalytic efficiency -proximity and orientation, covalent catalysis, acid-base catalysis, metal ion catalysis, strain and distortion theory. Mechanism of action of Lysozyme, Carboxypeptidase, Chymotrypsin and Ribonuclease. Reversible and Irreversible inhibition Competitive, Non-competitive and mixed inhibition. Substrate inhibition and Feedbackinhibition. Therapeutic, diagnostic and industrial applications of enzyme inhibitors

UNIT - IV **Hours 18**

INTRODUCTION OF CO-ENZYMES: Structure and functions Pyridine and flavin nucleotides, coenzyme A, Pyridoxal phosphate and thiamine pyrophosphate, tetrahydrofolate and B12 Coenzymes. Allosteric Interactions: Enzyme regulation, allosteric enzymes. Allosteric kinetics (MWC and KNF models), symmetry and sequential models. Hill's equation and Hill's coefficient. Isozymes.

UNIT - V **Hours 18**

MULTIENZYME SYSTEM: Multifunctional enzymes. Multi-enzyme complexes (Pyruvate dehydrogenase complex, fatty acid synthase and Na K ATPase), Metalloenzymes. Modern concepts of evolution of catalysts: Catalytic RNA (Ribozymes), abzymes. Immobilized enzymes and their industrial applications. Chemical modification and site-directed mutagenesis of enzymes. Industrial applications of enzymes food and pharmaceutical enzymes. Biosensors.

Reference Book:

1. Enzymes: Biochemistry, Biotechnology, Clinical Chemistry, 2ndedition, 2008 Trevor Palmer and Philip Bonner
2. Principles of Biochemistry, 1993. A.L. Lehninger, Nelson &Cox (CBS, India) and new edition.
3. Biochemistry, 2004, Donald Voet and Judith Voet, John Wiley and sons. ISBN-047119350
4. Biochemistry, 5th edition, by Lubert Stryer, New York: WH Freeman; 2002, ISBN-10: 0-7167-3051-0.
5. TextBook of Biochemistry with clinical correlations, 4th edition Thomas M. Devlin.
6. Text of Biochemistry, 1908-West & Todd, MacMillian Publications
7. Principles of Biochemistry, by G. L. Zubay, 1995, Wm. C. Brown
8. Biochemistry 2nd edition Christopher K.Mathews and K.E.VanHoldge (1995) (Benjamin and cumming).
9. Enzymes. Dixon and Webb 3rded. Longmans, 1979.
10. Enzymes in Food Technology. CRC Press, 2001. Whitehurst RJ..
11. Industrial Enzymes and their applications. Uhlig H. John Wiley, 1998. 12.

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FIRST YEAR	CORE COURSE - V	CREDIT : 5
SEMESTER II	MOLECULAR BIOLOGY	P25BCCC22

COURSE OUTCOMES:

- This course is about genes their structure and function therefore, students will study the mechanics of replication, repair, transcription, and translation in bacteria, archaea and eukaryotes,
- A central goal is to understand gene regulation at all levels, and the structure-function relationships of nucleic acids and proteins.

COURSE SPECIFIC OUTCOMES:

- The course gives deep insight into the molecular mechanisms behind the existence of life.
- Students will understand the structure and organization of genomes and its functions in lower to higher forms of life.
- Students will learn about the regulation of gene expression.
- Course covers the fundamental molecular causes behind the several non-communicable and communicable diseases. Which enables the students to pursue careers in healthcare and clinical research.

UNIT - I **Hours 18**

CENTRAL DOGMA OF LIFE: DNA as the genetic material, supercoiling, hybridization. Hierarchy of Chromatin Organisation, Central Dogma, Unique sequence DNA, Repetitive DNA SINES, LINEs, Satellite, Minisatellite and Microsatellite DNAs, C-Value Paradox. E.Coli Chromosome and plasmids, Mitochondrial and Chloroplast Genomes. Concept of genes. Structure of Protein-coding genes in prokaryotes and eukaryotes

UNIT - II **Hours 18**

DNA REPLICATION, REPAIR AND MUTATION:DNA in prokaryotes and eukaryotes. Mode of replication; experimental findings of Meselson &Stahl. Enzymes involved in replication, events on the replication fork and termination, mechanism of replication. Inhibitors of DNA replication and DNA repair mechanisms (Direct repair, excision repair, mismatch repair, recombination repair, SOS response, Eukaryotic repair system). Type of damages and mutation point mutation and frameshift mutation. Suppressor Gene

UNIT - III **Hours 18**

TRANSCRIPTION: Organization of transcriptional units. prokaryotes and eukaryotes. RNA polymerases structure and functions. Promoters, transcription factors, transcription complex assembly and mechanism of transcription- Transcriptional regulation hormonal (steroid hormone

receptors), phosphorylation (STAT proteins). Post-transcriptional processing. Alternative splicing. Catalytic RNA (ribozymes), antisense RNA. Inhibitors of transcription.

Hours 18

UNIT - IV

TRANSLATION: The specificity, redundancy and wobble hypothesis. Mitochondrial and chloroplast genetic codons. Components of protein synthesis mRNA, RNA and tRNA. Mechanism of protein synthesis. Regulation of protein synthesis. Constitutive and narrow domain regulation. Inhibition of protein synthesis. Co-and post-translational modifications. Protein targeting the signal sequence hypothesis, targeting proteins to membranes, nucleus and intracellular organelles. Protein degradation ubiquitin pathway. Protein folding models, molecular chaperones.

Hours 18

UNIT - V

GENE EXPRESSION AND REGULATION: Levels of gene expression. Principles of gene regulation. Upregulation, downregulation, induction and repression. Operon models Lac operon, Trp operon. Comparison of gene regulation strategies in prokaryotes and eukaryotes. Genetic and epigenetic gene regulation by DNA methylation. Methylation and gene regulation in mammals and plants. Chromosome remodelling –acetylation & deacetylation , role of transcription factor & hormone in eukaryotic transcription regulation.

Reference Book:

1. Enzymes: Biochemistry, Biotechnology and Clinical chemistry, 2nd edition, 2007,
2. Palmer T and Bonner P; Affiliated- East West press private Ltd, New Delhi
3. Fundamentals of Enzymology, 3rd edition, 2003, Price NC and Stevens L; Oxford University Press, New York
4. Voet's Biochemistry, Adapted ed, 2011,Voet,D and Voet JG; Wiley, India
5. Lehninger Principles of Biochemistry, 8th edition, 2021, .Nelson DL and Cox MM; WH Freeman & Co, New York
6. Biochemistry, Berg JM, Stryer L, Gatto,G, 8th ed, 2015;WH Freeman & Co., New York.
7. Enzyme Kinetics and Mechanism; Cook PF, Cleland W, ;2007; Garland Science,

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FIRST YEAR	CORE COURSE – VIII	CREDIT : 5
SEMESTER - II	METABOLISM & REGULATION	P25BCCC23

COURSE OUTCOMES:

- To learn the metabolism and integration of biomolecules that takes place in the human system.
- Integrate the various aspects of metabolism & their regulatory pathways.
- Students can understand the fundamental energetic of biochemical processes.
- To elaborate the relation between biochemical defects and metabolic disorders.
- To follow the organization of signaling pathways.

COURSE SPECIFIC OUTCOMES:

- To learn the metabolism and integration of biomolecules that takes place in human system.
- Integrate the various aspects of metabolism & their regulatory pathways
- Students can understand the fundamental energetics of biochemical Processes
- To understand the relation between biochemical defects and metabolicDisorders.
- To understand the organization of signaling pathways.
- To understand the role of membrane processes in metabolism
- Overall, gaining an understanding of the processes of metabolic transformation at the molecular level and how these processes are studied.
- Afford students opportunity to appreciate the relevance/applications of biochemistry in our daily activities.

Hours 18

UNIT - I

INTRODUCTION: Overview of major classes of biomolecules, forces stabilizing biomolecules. General scheme of metabolism, historical and experimental details in derivation of a metabolic pathway, catabolic, anabolic and amphibolic pathways. Oxidative phosphorylation. Electron transport chain. Standard free energy change of a chemical reaction, redox potentials, ATP and high energy phosphate compounds.

Hours 18

UNIT – II

CARBOHYDRATES METABOLISM: Glycolysis and gluconeogenesis pathway, key enzymes and co-ordinate regulation. The citric acid cycle and its regulation. The pentose phosphate pathway. Metabolism of glycogen and regulation. Key junctions in metabolism-glucose-6-phosphate, pyruvate and acetyl CoA. Blood glucose homeostasis role of tissues and hormones.

Hours 18

UNIT - III

AMINO ACID METABOLISM: Biosynthesis and degradation of amino acids and their Transamination and deamination, ammonia formation, the urea cycle and regulation of ureogenesis.

Hours 18

UNIT - IV

LIPIDS METABOLISM

Lipogenesis: Biosynthesis of fatty acid, triglycerides, phospholipids, and cholesterol. Regulation of triacylglycerol, phospholipids and cholesterol biosynthesis. Oxidation of lipids. Role of carnitine cycle in the regulation of B oxidation. Cholesterol Catabolism, Ketogenesis and its control. Lipoprotein metabolism exogenous and endogenous pathways

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Hours 18

UNIT - V

NUCLEIC ACID METABOLISM:

Biosynthesis and catabolism of purines and pyrimidines and their regulation. (https://www.slideshare.net/senchiy/nucleic-acids-and-nucleotide)

Reference BooK:

1. Biochemistry L. Stryer.
2. Principles of Biochemistry R. Horton et al.
3. Lehninger's Principles of Biochemistry D. L. Nelson and M. M. Cox.
4. Harpers Biochemistry R. K. Murray et al.
5. Principles of Biochemistry- G. L. Zubay.
6. Outlines of biochemistry- Lippincott

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FIRST YEAR	CORE PRACTICAL - II	CREDIT : 3
SEMESTER - II	ENZYMOLOGY & MOLECULAR BIOLOGY(P)	P25BCCP2

COURSE OUTCOMES:

- To inculcate skill in students enabling them to apprehend the wider knowledge about principles and techniques to be employed forth essay of enzymes under investigation.
- To inculcate the knowledge of isolation and purification techniques of enzymes using alkaline phosphataseas an example
- To perform experiments to study the factors affecting enzyme activity
- To achieve raining in assay of enzymes

COURSE SPECIFIC OUTCOMES:

- The student will be able to employ the relevant techniques for isolation and purification of enzymes and gain skill in kinetic studies which is essential for research activity
- Stadent will acquire ability in performing enzyme assay, and explicate the methods that lotin the basis of enzyme characterization

HOURS : 90

Practicals

1. Isolation of Alkaline Phophatase from goat kidney.
2. Purification of alkaline phosphatase
3. Checking the purity using SDS-PAGE
4. Determination of optimum pH and temperature of alkaline phosphatase.
5. Determination of specific activity and Km of alkaline phosphatase.
6. Effect of activators and inhibitors on the activity of alkaline phosphatase.
7. Isolation of genomic DNA (Goat liver, yeast)
8. Isolation of RNA
9. Separation of DNA by agarose gel electrophoresis

Reference Book:

1. David Plummer (2001) An Introduction to Practical Biochemistry (3rd ed) McGraw Hill Education (India) Private Ltd
2. Jayaraman, J (2011), laboratory Manual in Biochemistry, New age publishers
3. Fundamentals of Enzymology; 3rd Edn. Nicholas C. Price and Lewis Stevens, Oxford University Press (2012).
4. Enzymes: A Practical Introduction to Structure, Mechanism, and Data Analysis; Robert Copeland , Wiley-VCH Publishers (2000).

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FIRST YEAR	ELECTIVE COURSE – II A	CREDIT : 4
SEMESTER - II	BIOINFORMATICS	P25BCE2A

COURSE OUTCOMES:

- The purpose of studying this paper is to apply computational facilities in different fields of life sciences, physical and chemical sciences.
- After completion, students could learn drug designing through computer based modification programs using synthetic or natural sources.
- Most important application of Bioinformatics is in the field of drug discovery where it reduces more than 60% of the time, money and labor.

COURSE SPECIFIC OUTCOMES:

- To get introduced to the basic concepts of Bioinformatics and its significance in Biological data analysis.
- Explain about the methods to characterize and manage the different types of Biological data.
- Introduction to the basics of sequence alignment and analysis,

UNIT - I **Hours 12**

BIOINFORMATICS: An overview, Definition & History; Bioinformatics databases & tools on the Internet- NCBI, EBI, PIR, Swiss-Prot, GenBank pattern and motif searches-BLOCKS, PRINTS, PFAM

UNIT - II **Hours 12**

PROTEINS AMINO ACIDS:Levels of protein structure Ramachandran Plot. Protein Secondary structure prediction Chou-Fasman rules, Gamier-Osguthorpe-Robson (GOR) methods; Predicting 3D structure homology modeling, threading fold recognition and ab initio methods Rosetta CASP

UNIT - III **Hours 12**

BIOLOGICAL SEQUENCE ANALYSIS: Pairwise sequence comparison Sequence queries against biological databases BLAST and FASTA Multiple sequence alignments - Phylogenetic alignment. Algorithms and Matrices: Scoring matrices- PAM and BLOSUM; dynamic programming Algorithms, Needleman and Wunsch, Smith-Waterman;

UNIT - IV **Hours 12**

.PROTEIN STRUCTURE VISUALIZATION TOOLS: Ras Mol, HEX, Argus Lab Swiss PDB Viewer alignment and analysis SCOP, CATH, FSSP, UNIX. Structure Classification, alignment and analysis SCOP, CATH, FSSP, UNIX.

UNIT - V **Hours 12**

FUNCTIONAL GENOMICS (METABOLISM AND REGULATION) IN BIOCHEMISTRY:Sequencing genomes- Genome databases on the web, Prokaryotic Genome

Database with comparison with Human genome, HGP, GENE CLUSTER, DNA Microarray, SWISS-2DPAGE Database, TIGR, WIT, CYTOSCAPE and DRUG DISCOVERY

Reference Book:

1. Bioinformatics-Sequence and Genome Analysis- David W. Mount, Cold Spring Harbor Laboratory Press (2004).
2. Introduction to Bioinformatics, Attwood, T.K. and D.J. Parry-Smith, Pearson Education Ltd., New Delhi (2004).
3. Bioinformatics Westhead, D.R., Paris J.H. And R.M. Twyman, Instant Notes: Viva Books Private Ltd, New Delhi (2003).
4. Introduction to Bioinformatics, Arthur M. Lesk, Oxford University Press, New Delhi (2003).
5. Bioinformatics- Sequence, structure and databanks, Higgins D. and W. Taylor (Eds), Oxford University Press, New Delhi (2000).
6. Bioinformatics; A practical Guide to the Analysis of Genes and Proteins, Wiley-Interscience,Baxevanis, A. and B.F. Ouellette, Hoboken, NJ (1998).
7. Introduction to computational Biology, Michael, S. Waterman, Chapman & Hall, (1)

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FIRST YEAR	ELECTIVE COURSE – II B	CREDIT : 4
SEMESTER II	NANOTECHNOLOGY	P25BCE2B

COURSE OUTCOMES:

- Understand the principles of drug delivery systems and control of varied parameters for effective drug delivery.
- Gain knowledge on the mode of action of nanoparticle activity inside the cellular structure..
- Inculcate the concepts of advances in Nano therapeutics

COURSE SPECIFIC OUTCOMES :

- Remember the elemental principles of drug delivery systems
- Understand the mode of action of nanoparticles and it's in vitro toxicity assays
- Recent developments and understanding the available therapy in cancer treatment
- Understanding of most recent advances in Nanobiotechnology with novel Techniques.

UNIT - I **Hours 12**

PRINCIPLES OF DRUG DELIVERY SYSTEMS: Modes of drug delivery, Absorption distribution metabolism excretion characteristics of drugs, Controlled drug delivery site specific drugs, Barriers for drug targeting passive and active targeting, Strategies for site specific drug

UNIT - II **Hours 12**

TOXICITY ASSAYS AND THEIR PRINCIPLES: Cell viability, LDH release, ROS production, Morphological observation, Membrane potential, Live/Dead assay, Comet Assay, Cell cycle analysis and Apoptosis detection by flow cytometer

UNIT - III **Hours 12**

NANOPARTICLES AND CANCER THERAPY: Cancer and its types: Mechanisms of progression in Cancer: Cellular trafficking, Cancer invasion, Migration, Angiogenesis and Metastasis. Chemotherapy, Immunotherapy, Photodynamic Therapy (PDT), Photothermal Therapy (PTT), Magnetic Hyperthermia (MHT), High Intensity Focused Ultrasound (HIFU).

UNIT - IV **Hours 12**

TARGETED DRUG DELIVERY: Classification of targeted drug delivery systems, Bioconjugation, Nanoparticles surface modification PEGylation, Gold nanoparticles for drug delivery, Magnetic nanoparticles as drug

UNIT - V **Hours 12**

BIO-PRINTING (THREE DIMENSIONAL BIOPRINTING):Introduction,History, principle and its components, Classification of 3D bioprinting techniques Extrusion-based bioprinting, Droplet-based

bioprinting, Laser-based bioprinting. Design Requirements for 3D Bioprinting- Magnetic Resonance Imaging, Computed Tomography, Computer-Aided Design Based Systems, 3D modeling softwares, Bio inks for 3D bioprinting Applications of 3D Bioprinting and future trends.

Reference Book:

1. Drug delivery: Principles & applications Wang, B., Hu, L, Siahaan, T.J, John Wiley& Sons, (2016).
2. 3D Bio-printing-Fundamentals, Principles and Applications, Ibrahim T. Ozbolat, Academic Press, (2016).
3. 3D Bio-printing in Regenerative Engineering, Principles and Applications, Ali Khademhosseini, Gulden Camci-Unal, 1st edition, CRC press, (2018).

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FIRST YEAR	NON – MAJOR ELECTIVE COURSE – I A	CREDIT : 2
SEMESTER - II	NUTRITIONAL BIOCHEMISTRY	P25BCNM1A

COURSE OUTCOMES

- To understand basic concepts involved in growth , health, nutrition, physiology and metabolism
- To discuss the concepts and applications of nutrition in correlation with biochemistry
- To define nutritional needs in healthy individuals and modification of diet during illness.

COURSE SPECIFIC OUTCOMES

- Capable of describing biochemical pathways relevant in nutrient metabolism
- Capable of describing biochemical techniques that are relevant for the investigation of the Nutrient metabolism.
- To introduce latest concept of dietary Management.
- Describes the metabolic functions and deficiency of micro & macro nutrients

UNIT - I **Hours 6**

Basic concepts - Nutrition - Food groups and balanced diet. Novel Foods. Calorific value of foods: Direct and indirect calorimetry. Empty calories. Basal metabolic rate: Factors affecting BMR. SDA and physical activity. Calculation of day's energy requirement. Assessment of nutritional status. Lactose intolerance. Nutritional requirement and biochemical changes in different physiological states -infancy, childhood, pregnancy, lactation, and ageing. Sports

UNIT - II **Hours 6**

Elements of nutrition - Plant and animal sources of simple and complex carbohydrates, fats and proteins and their requirement. Biological significance, deficiency and toxicity of macronutrients and micronutrients. Role of dietary fibre. Protein sparing action of carbohydrates and fats. Essential amino acids. Essential fatty acids. Effects of naturally occurring food toxins, preservatives, additives, alcohol and tobacco on health.

UNIT - III **Hours 6**

Vitamins and Minerals- Dietary sources, classification, biochemical functions, requirements, absorption, metabolism and excretion. Vitamin B complex as coenzyme. Nutritional significance of dietary calcium, phosphorus, magnesium, iron, iodine, zinc and copper.

UNIT - IV **Hours 6**

Malnutrition - Diseases arising due to Protein - Calorie Malnutrition and undernutrition (Kwashiorkor and Marasmus), Prevention of malnutrition. Deficiency diseases associated with

vitamin B complex, vitamin C and A, D, E & K vitamins - Mineral deficiency diseases - aetiology, sign and symptoms and dietary supplementation. Enrichment and fortification (vitamins and minerals)

UNIT - V **Hours 6**

Nutrition in diseases - Aetiology, signs and symptoms , treatment and dietary management during fever(Typhoid and Malaria) and infectious diseases(COVID-19), Jaundice, hyper acidity (Ulcer), Atherosclerosis, Hypertension, kidney diseases and diabetes in adults. Starvation and Obesity. Inter-relationship of nutrition, infection, immunity and poverty.

Reference BooK:

1. Srilakshmi. E .(2016) Nutrition Science, New Age International Publishers.
2. Mahan, Kathleen L. (2004) Krause's Food, Nutrition and Diet Therapy, W.B.Saunder's 11th Edition
3. Andreas M. Papas (1998). Antioxidant Status, Diet, Nutrition, and Health (1st ed) CRC Press.
4. M. Swaminathan (1995) Principles of Nutrition and Dietetics. Bappco
5. Margaret Mc Williams (2012). Food Fundamentals (10th ed) Prentice Hall

Tom Brody (1998) Nutritional Biochemistry (2nd ed). Academic Press, USA

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FIRST YEAR	NON – MAJOR ELECTIVE COURSE – I B	CREDIT : 2
SEMESTER - II	HERBAL MEDICINE	P25BCNM1B

COURSE OUTCOMES :

- The subject includes fundamentals of plant taxonomy, plant parts, classification of plants; study of important plant families; cultivation of medicinal plants, harvesting and storage, etc.

COURSE SPECIFIC OUTCOMES

- Capable of describing medicinal plants relevant in pharmocology
- Capable of describing medicinal plants that are relevant for the investigation of the Drug preparation.
- To introduce latest concept of pharmacological studies using medicinal plants.
- Describes the Systems of Indian Medicines

UNIT - I **Hours 6**

Systems of Indian Medicines: Ayurveda, Unani, Siddha and Homeopathy. Classification of Crude drugs: Morphological, Taxonomical, Chemical and Pharmacological

.UNIT - II **Hours 6**

Morphology and general anatomy of medicinally important plant parts: Roots, Stem and its modifications, Barks, Leaves, Flowers, Fruits, Seeds.

UNIT - III **Hours 6**

Taxonomy: Elementary knowledge of Binomial nomenclature, Outline of Bentham and Hooker classification, Preparation of herbarium specimens.

UNIT - IV **Hours 6**

Study of some medicinally important families (diagnostic features with at least three examples of species of medicinal use): Paparveraceae, Rutaceae, Fabaceae, Apiaceae,Rubiaceae, Asteraceae, Solanaceae, Scrophulariaceae, Lamiaceae, Liliaceae.

UNIT - V **Hours 6**

Cultivation methods, Herbal pesticides, Plant growth regulators. Harvesting and Storage. Marketing and general aspects of export of medicinally important plants.

Reference BooK:

1. A Class Book of Botany. A.C. Dutta. Oxford University Press

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SEMESTER – III

SECOND YEAR	CORE COURSE - VII	CREDIT : 5
SEMESTER - III	IMMUNOLOGY	P25BCCC31

COURSE OUTCOMES

The objective of this course are to

- Introduce the structure and functions of lymphoid organs and cells of the immune system.
- Illustrate the structure and classification of antibodies and adaptive immune response.
- Impart knowledge on the types of immunity and uses of vaccines.
- Provide an understanding of immune related diseases and transplantation.
- Study the Ag-Ab interaction and immunological techniques to identify antigens and antibodies.

COURSE SPECIFIC OUTCOMES

- Associate structure and function of the organs involved in our body's natural Defence.
- Classify antigens and antibodies and the role of lymphocytes in defending the host.
- Describe the types of immunity and the uses of vaccines.
- Understand the immune related diseases and mechanism of transplantation.
- Examine the immunological tests and relate it to the immune status of an Individual.

UNIT I **Hours 18**

Elements of Immunology. Types of immunity- innate and acquired. Humoral and cell mediated immunity. Central and peripheral lymphoid organs- Thymus, bone marrow, spleen, lymph nodes and other peripheral lymphoid tissues- GALT. Cells of the immune system- lymphocytes, mononuclear phagocytes- dendritic cells, granulocytes, NK cells and mast cells, cytokines. Antigens vs immunogens, Haptens. Factors influencing immunogenicity. Immunoglobulins structure, classification and functions. Isotypes, allotypes and idiotypes.

UNIT II **Hours 18**

Complement activation and its biological consequences. Clonal selection theory. Organization and expression of immunoglobulin genes generation of antibody diversity. Class switching. T-cell, B-cell receptors, Antigen recognition- processing and presentation to T cells. Interaction of T and B cells. Immunological memory. Effector mechanisms- macrophage activation. immunotolerance, immunosuppression.

UNIT III **Hours 18**

Cell mediated cytotoxicity, MHC genes and products. Polymorphism of MHC genes, role of MHC antigens in immune response, MHC antigens in transplantation. Transplantation types. Immune responses to infectious diseases- Viral, bacterial and protozoal. Cancer and immune system. AIDS and other immunodeficiency disorders. Autoimmunity. Hypersensitivity- types.

UNIT IV **Hours 18**

Immunization practices- active and passive immunization. Vaccines- killed, attenuated-toxoids. Recombinant vector vaccines- DNA vaccines, synthetic peptide vaccines- anti idiotype vaccines production of polyclonal and monoclonal antibodies. Principles, techniques and application. Genetically engineered antibodies. Fractionation of leucocytes by density gradient centrifugation. Identification of lymphocytes and their subsets in blood. Leukocyte migration inhibition technique. Delayed type hypersensitivity technique

UNIT V **Hours 18**

Agglutination and precipitation techniques. Immuno-electrophoresis, RIA, immunoblotting, Avidin- biotin mediated immuno assay. Immunohistochemistry-immunofluorescence, immunoferritin technique. Fluorescent immunoassay. Cytokines assay: ELISA and ELISPOT. Production of cytokines in vitro. Interferon production. Abzymes. Experimental animal models: inbred strains, SCID mice, nude mice, knock out mice cell culture system: Primary lymphoid culture, cloned lymphoid cell lines.

REFERENCE BOOK

- Roitt Roitt's et al. (2001). Essential Immunology. 10th ed. Blackwell Sci.
- Richard A. Goldsby et al (2003). Kuby Immunology. 4th ed. WH Freeman & Co.
- Abbas et al., (2000). Cellular and Molecular Immunology. W.B. Saunders Company.
- Janeway C (Ed), (2001). Paul Travers. Immunobiology. 5th ed. Garland Publ.
- Eli Benjamini AU et al. (2000). Immunology: A short course. 4th ed. Wiley-Liss.
- NMS Series in Immunology. 3rd ed. Lippincott Willams & Wilkins.
- Bier et al. (1986). Fundamentals of immunology Springer Verlag.

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SECOND YEAR	CORE COURSE - VIII	CREDIT : 5
SEMESTER - III	RESEARCH METHODOLOGY	P25BCCC32

COURSE OUTCOMES

The objectives of the course are to:

- Introduce the components of research.
- Acquaint on the experimental design and literature survey.
- Analyse the data and find out the significance statistically.
- Highlight the importance of computation in research.
- Provide mechanics of writing a research report and-on experience in designing and working on small projects.

COURSE SPECIFIC OUTCOMES

- Explain the types of research and formulate and plan the research.
- Design experimental setup, review the literature, compile and document the data.
- Analyze and validate the experimental data using statistical tools.
- Interpret the data using computational tools.
- Compile and draft a research report, present results findings and publish ethically.

UNIT - I

Hours: 18

Characteristics and types of Research, Research Methods versus Methodology, Research designs in Biochemistry: experimental, *in vitro, in vivo, in situ*, clinical trials. Identification and criteria of selecting a research problem (Hypothesis); Formulation of objectives; Research plan and its components.

UNIT - II

Hours: 18

Experimental design - Objective, Design of work, Guidelines for design of experiments, Literature Search - Databases for literature search, Material and methods, Designing biological experiments, Compilation and documentation of data.

UNIT - III

Hours: 18

Statistical Analysis: Measures of variation - standard deviation, Non-linear regression, Standard error. Analysis of variance for one-way and two-way classified data and multiple comparison procedures. Significance - students "t" test, chi-square test, Dunnet's test.

UNIT - IV

Hours: 18

Computer and its role in research: Basics of MS word, MS Excel: tabulation, calculation and data analysis, preparation of graphs, histograms and charts. Use of statistical software SPSS. Power Point: preparation of presentations and scientific poster designing.

UNIT - V **Hours: 18**

Scientific writing for journals - Preparation of Abstract, Impact factor, h-index, i-10 index, citation index, Dissertation/Thesis writing: format, content and chapterization, writing style, drafting titles & sub-titles, captions and legends. Writing results, discussion and conclusions. Bibliography and references, referencing style - Harvard and Vancouver systems, Appendices and acknowledgement; Ethical issues in research; Intellectual property right and plagiarism.

REFERENCE BOOKS

- Garg BL, Karadia R, Agarwal F and Agarwal UK (2002). An introduction to Research Methodology, RBSA Publishers.
- Kothari CR (2004). Research Methodology: Methods and Techniques. New Age International.
- Sinha SC and Dhiman AK (2002). Research Methodology, Ess Ess Publications. 2 volumes.
- Gurumani N (2014). Research Methodology for biological Sciences, MJP Publishers.
- Prabhat Pandey and Meenu Mishra Pandey (2015). Research Methodology: Tools and Techniques.
- Coley SM and Scheinberg CA (1990). "Proposal Writing", Sage Publications.
- Day RA (1992). How to Write and Publish a Scientific Paper, Cambridge University Press.
- Fink A (2009). Conducting Research Literature Reviews: From the Internet to Paper. Sage Publications.
- Scientific Thesis Writing and Paper Presentation (2010). MJP Publishers.
- Suresh C Sinha and Anil K. Dhiman (2002). Research Methodology (2 Vols-Set), Vedams Books (P) Ltd.

WEB RESOURCES

- https://explorable.com/research-methodology
- http://www.scribbr.com
- http://www.open.edu
- http://www.macmillan.ihe.com.

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SECOND YEAR	CORE COURSES - IX	CREDIT : 5
SEMESTER - III	CLINICAL BIOCHEMISTRY	P25BCCC33

COURSE OUTCOMES

The main objectives of this course are to

- To understand the need and methods of various biological sample collection.
- To explicitly understand the etiopathogenesis, symptoms and complications of metabolic and hormonal disorders and the relevant diagnostic markers
- To emphasize the diagnostic significance of serum enzymes in different pathologies and other Laboratory investigations of diagnostic importance so as to
- differentiate normal from disease

COURSE SPECIFIC OUTCOMES

- To conceive the role of inherited genes in inborn errors of metabolism and
- methodologies pertaining to in utero diagnosis and post-natal screening.
- To get updated about electrolyte and hormonal imbalances and the biochemical tests to
- diagnose them.

UNIT - I

Hours 18

Biochemical investigations in diagnosis, prognosis, monitoring, screening: Specimen collection – blood, (primary /Secondary specimen)., urine and CSF. Preservation of biological specimens -blood, urine, CSF and amniotic fluid; Biological reference ranges; Disorders of blood cells: Hemolytic, iron deficiency and aplasticanemia and diagnosis, sickle cell anaemia, thalassemia HBA1C variants. Porphyrias, Thrombocytopenia, Causes of leucopenia, leukemia and leucocytosis. Disorders of blood clotting mechanism – Von willebrand's disease, Hemophilia A, B and C, diagnostic test for clotting disorders, D-dimer and its clinical significance.

UNIT - II

Hours 18

Diabetes mellitus: pathology and complications: Acute changes; Chronic complications: Diabetic nephropathy, neuropathy, retinopathy and Diabetic foot ulcers, Random/Fasting/PP glucose testing, Impaired glucose tolerance (IGT), Impaired fasting glucose (IFT), Diagnosis-by GTT, Pre-diabetes, Gestational DM, Glycosylated Haemoglobin (HBA1c); Glycated albumin., Hypoglycaemia and critical alert value for glucose. Markers of complications of Diabetes mellitus: Metabolic syndrome, Lipid profile &lipoproteinemia, Atherosclerosis, Diabetic nephropathy, Micralbuminuira, eGFR. Point of care testing for glucose (Glucometers) and continuous glucose monitoring (CGM): principle and its use. Major groups of anti-diabetic drugs. Diet and life style modifications.

UNIT - III

Hours 18

Diagnostic Enzymology: Clinically Important Enzymes and Isoezyme as diagnostic markers: Clinical significance of AST, ALT, ALP, ACP, CK, γ-GT, amylase,

pseudocholinesterase and their pattern in. Myocardial infarction; Liver disease, Bone disease, Muscle disease, Cancer (tumor markers), GI tract pancreatitis); Enzymes as therapeutic agents. Pre- and post-natal testing: Amniocentesis, prenatal detection of inborn errors of metabolism in developing fetus- Autosomal recessive mode of inheritance- cystic fibrosis, X linked recessive inheritance-Duchenne muscular dystrophy. New born screening (NBS) for Inborn errors of metabolism, Tandem mass spectrometry application in NBS.

UNIT - IV **Hours 18**

Liver function tests: Liver function test panel, Fatty liver. Plasma protein changes in liver diseases. Hepatitis A, B and C. Cirrhosis and fibrosis. Portal hypertension and hepatic coma. Acute phase proteins -CRP, Haptoglobins, α-fetoprotein, ferritin and transferrin and their clinical significance, Interpreting serum protein electrophoresis. Inflammatory markers (cytokines such as TNF-alpha IL6 and others).

UNIT - V **Hours 18**

Renal function tests - tests for glomerular and tubular Function-Acute and chronic renal Failure-Glomerulonephritis, Nephrotic syndrome, uraemia-urinary calculi-Nephrocalcinosis and Nephrolithiasis-causes, pathology and symptoms. Chronic kidney disease. Dialysis-Hemodialysis and peritoneal dialysis.

REFERENCE BOOK

- Thomas M. Devlin (2014). Textbook of Biochemistry with Clinical Correlations (7th ed). John Wiley & Sons
- Montgomery R, Conway TW and Spector AA (1996). Biochemistry: A Case-Oriented Approach (6th ed), Mosby Publishers, USA.
- Saunders Tietz (2018). Fundamentals of Clinical Chemistry and Molecular Diagnostics (8th ed).
- Dinesh Puri, (2020) Text book of Biochemistry: A clinically oriented approach – 4th
- Edition, Elsevier.
- Chatterjee MN and Rana Shinde (2012). Textbook of Medical Biochemistry (8th ed),
- Jaypee Brothers Medical Publishers.
- Poonam Agrawal (2021). Clinical Case Discussion In Biochemistry A Book On Early Clinical Exposure (ECE), CBS Publishers & distributors pvt. Ltd.

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SECOND YEAR	CORE PRACTICAL - III	CREDIT : 3
SEMESTER - III	CLINICAL BIOCHEMISTRY AND IMMUNOLOGY (P)	P25BCCP3

COURSE OUTCOMES

The objectives of this course are to

- Introduce the methods of sample collection (blood & urine) for analytical purpose.
- Impart practical knowledge on the assay of activity of various diagnostically important enzymes
- Understand the estimation procedure for various important biomolecules.
- Help students learn the routine qualitative analysis of urine sample for diagnostic purpose.
- Train students on various hematological tests and its significance.

COURSE SPECIFIC OUTCOMES

- Acquaint knowledge on collection of biological samples (urine, blood) and their preparation for diagnostic purpose.
- Assay the activity of various clinically important enzymes and relate their clinical importance.
- Estimate the important biomolecules in biological samples and relate their clinical significance
- Qualitatively analyze urine sample for normal and abnormal constituents in urine and interpret the results
- Perform the routine haematological tests.

Hours 90

1. Estimation of creatinine by Jaffe's method (serum &urine).
2. Estimation of urea by diacetyl monoxime method (serum & urine).
3. Estimation of uric acid (serum & urine).
4. Estimation of cholesterol by Zak's method.
5. Estimation of Glucose by Ortho Toluidine method.
6. Estimation of Protein by Lowry's method.
7. Estimation of Hemoglobin by Shali's/Drabkins method.
8. Antibody titer by ELISA method.
9. Precipitin reaction by double immunodiffusion and radial immunodiffusion (Ouchterlony and Mancini's methods).
10. Seperation of mononuclear cells by Ficoll – Hypaque.
11. Widal Test.
12. Rocket Immuno Electrophoresis.
13. Qualitative analysis of normal constituents of urine

Urea, Creatinine, Phosphorus, Calcium
Abnormal constituents

a) Calcium
b) Sugar (Glucose, fructose, pentose)
c) Protein

d) Aminoacids (Tyrosine, Histidine, Tryptophan)
e) Ketone bodies
f) Bile pigments with clinical significance.

REFERENCE BOOKS

- Manickam SS (2018). Biochemical Methods (3rd ed.). New age International Pvt Ltd publishers - ISBN 10: 8122421407 / ISBN 13: 9788122421408.
- Plummer DT (n.d.). An Introduction to Practical Biochemistry. Tata Mc Graw Hill- ISBN: 97800708416.
- Alan H Gowenlock (1998). Varley's Practical Clinical Biochemistry, 6th edition, CBS Publishers, India.
- Godkar B (2020). Textbook of Medical Laboratory Technology Vol 1 & 2 Paperback, 3rd edition, Bhalani Publishers.
- Kanai L Mukerjee (1996). Medical Lab Technology, Vol I& II, 1st edition, Tata Mcgraw Hill, Pennsylvania.
- Ranjna Chawla (2014). Practical Clinical Biochemistry Methods and interpretations 58 (Paperback). 4th edition, Jaypee Brothers Medical Publishers, New York.
- Singh SK (2005). Introductory Practical Biochemistry (2nd ed.). Alpha Science International, Ltd- ISBN 10: 8173193029 / ISBN 13: 9788173193026.
- Ashwood BA (2001). Tietz Fundamentals of Clinical chemistry. WB Saunders Company, Oxford Science Publications USA - ISBN 10: 0721686346 / ISBN 13: 978072168634.
- Talwar GP and Gupta SK. Practical and clinical immunology. 2nd edition.
- Asim Kumar Roy. Immunology Theory and Practical.
- Hay C and Owyn MR. Practical Immunoogy. Westwood, 4th edition.
- SuhadHadi Mohammed, MdMohsin Ahmed, Practical Immunology.
- Christine Dorreteyn Steven, 3rd Edition, Clinical Immunology Serology.

WEB RESOURCES

- https://www.elsevier.com/journals/clinical-biochemistry/0009-9120/guide-for-authors.
- http://rajswasthya.nic.in/RHSDP%20Training%20Modules/Lab.%20Tech/Biochem stry/ Dr.%20Jagarti%20Jha/Techniques%20In%20Biochemistry%20Lab.pdf.
- https://dspace.cuni.cz/bitstream/handle/20.500.11956/111493/Clinical_biochemistrypd f.pdf?sequence=1&isAllowed=y.
- https://dspace.cuni.cz/bitstream/handle/20.500.11956/111493/Clinical_biochemistrypd f.pdf?sequence=1&isAllowed=y.

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SECOND YEAR	ELECTIVE COURSE - III	CREDIT : 4
SEMESTER - III	GENOMICS & PROTEOMICS	P25BCE1A

COURSE OUTCOMES

- The objective of the course is to provide students with a comprehensive and concise overview of technologies pertinent to Genomics and Proteomics, their applications and demonstrate skills to apply the knowledge in scientific queries.
- This course will help the engineering students to appreciate the surplus value of combining data from different omics-applications as a systems approach.

COURSE SPECIFIC OUTCOMES

On successful completion of the course, the student will be able to:

- Explain the current genomics and proteomics technologies and exploit the same in the growing field of omics.
- Interpret data obtained through high throughput expression studies.
- Apply the computational skills to plan and execute a biomedical 'omics' project.

UNIT - I — Hours 12

Basics of Genomics and Proteomics: Brief overview of prokaryotic and eukaryotic genome organization; extrachromosomal DNA: bacterial plasmids, mitochondria and chloroplast.

UNIT - II — Hours 12

Genome Mapping: Genetic and physical maps; markers for genetic mapping; methods and techniques used for gene mapping, physical mapping, linkage analysis, cytogenetic techniques, FISH technique in gene mapping, somatic cell hybridization, radiation hybrid maps, in situ hybridization, comparative gene mapping.

UNIT - III — Hours 12

Genome Sequencing Projects: Human Genome Project, genome sequencing projects for microbes, plants and animals, accessing and retrieving genome project information from the web.

UNIT - IV — Hours 12

Comparative Genomics: Identification and classification of organisms using molecular markers- 16S rRNA typing/sequencing, SNPs; use of genomes to understand evolution of eukaryotes, track emerging diseases and design new drugs; determining gene location in genome sequence.

UNIT - V **Hours 12**

Proteomics: Aims, strategies and challenges in proteomics; proteomics technologies: 2DPAGE, isoelectric focusing, mass spectrometry, MALDI-TOF, yeast 2-hybrid system, proteome databases.

REFERENCE BOOK

- Primrose SB, Twyman RM, Primrose SB and Primrose SB (2006). Principles of Gene Manipulation and Genomics. Malden, MA: Blackwell Pub.
- Liebler DC (2002). Introduction to Proteomics: Tools for the New Biology. Totowa, NJ: Humana Press.

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SECOND YEAR	ELECTIVE COURSE - III	CREDIT : 4
SEMESTER - III	BIOETHICS AND IPR	P25BCE1B

COURSE OUTCOMES

To create an awareness of ethics associated with used of genetically modified organisms/cells and its rationale for use in living organisms.

- The students will understand the concepts of Bioethics and Biosafety.
- The students will realize the impact of Gene cloning in societal problems and also understand the need of the Bioethics.

COURSE SPECIFIC OUTCOMES

- To implicate the importance of protecting the scientific intellect by filing patent and understand the various offices for filing and maintaining patents.
- To understand the scope of patenting in biological research

UNIT - I **Hours 12**

Introduction to bioethics, human genome project and its ethical issues, genetic manipulations and their ethical issues, ethical issues in GMOs, foods and crops in developed and developing countries, environmental release of GMOs, ethical issues involved in stem cell research and use, use of animals in research experiments, animal cloning, human cloning and their ethical aspects, testing of drugs on human volunteers.

UNIT - II **Hours 12**

Impact of gene cloning& Bioethics-Issues concerning reproduction, Birth, life and Death (Artificial insemination, egg donation, IVF, embryo transplants, Prenatal diagnosis and sex selection & Abortion).

UNIT - III **Hours 12**

Human Rights: Definition, Classification and Scope of Human Rights. United Nations Commission for Human Rights, National and State Human Rights Commission. Article 21 of Indian Constitution – UDHR. Social issues of Human rights.

UNIT - IV **Hours 12**

Intellectual Property Rights (IPR): Introduction to patents, types, process involved in patenting in India, trademarks, copyright, industrial design, trade secrets, traditional knowledge, geographical indications, history of national and international treaties and conventions on patents, WTO, GATT, WIPO, Budapest Treaty, Patent Cooperation Treaty (PCT) and TRIPS. Patent databases: Searching international databases; analysis and report formation. Indian Patent Act 1970; recent amendments; filing of a patent application; precautions before patenting disclosure/non-disclosure; procedure for filing a PCT application. The patentability of microorganisms-claims, Characterization and repeatability disposition in the culture collections, legal protection for plants and other higher organisms, new plant varieties by rights, tissue culture protocols.

UNIT - V **Hours 12**

Patent filing and infringement: Patent application- forms and guidelines, fee structure, time frames; types of patent applications: provisional and complete specifications; PCT and convention patent applications, International patenting-requirement, financial assistance for patenting introduction to existing schemes; Publication of patents-gazette of India, status in Europe and US. Research Patenting: Patenting by researchers and Scientists-University/organizational rules in India and abroad. Detailed information on patenting biological products, Case studies on patents (basmati rice, turmeric, neem etc.), and patent infringement.

REFERENCE BOOK

- Ignacimuthu S (2009). Bioethics, Narosa Publication house, ISBN: 978-81-7319-966-0.
- SreeKrishna V (2007). Bioethics and Biosafety in Biotechnology, (1st ed.), New Age International Private Limited.
- Trayer PC, Fredrick R and Koch M (2002), Biosafety.
- Michigan State University Biosafety in Microbiological and Biomedical Laboratories, (2020) 6th Ed.
- Kankanala C (2007). Genetic Patent Law & Strategy, 1st Edition, Manupatra Information Solution Pvt. Ltd.,

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SECOND YEAR	NON MAJOR ELECTIVE - II	CREDIT : 2
SEMESTER - III	DETECTION METHODS OF FOOD ADULTERATION	P25BCNM2A

COURSE OUTCOMES

The objectives of this course are to

- Provide the key knowledge base and laboratory resources to prepare students for careers as professionals in various industries and research institutions.
- To educate students about common food adulterants and their detection.
- To introduce students to food safety and standardization acts.
- To make students skilled in food adulteration testing and analytical instrumentation.

COURSE SPECIFIC OUTCOMES

After successful completion of the course, students will be able to

- The adulteration of common foods and their adverse impact on health
- Comprehend certain basic skills of detecting adulteration in common foods.
- Extend their knowledge of detecting other kinds of adulteration.

UNIT - I

Hours 6

Testing Adulteration of Milk: Test A: Adulteration of Milk Physical Tests: Detergent Test Filter Test Flow. Test B: Chemical Tests: Clot on boiling test. Test for starch in Milk Test for cane sugar in Milk Test for Buffaloes Milk in Cow"s Milk Test for added colors in Milk Test for skim milk powder in milk Detects the presence of added carbonates and bicarbonates in milk Test for soda in milk. Test for glucose.

UNIT - II

Hours 6

Adulteration of Ghee: Test for vegetable fat: Nitric acid test Soda ash test Valenta test Test for added alkali. Baudouin test. Analysis of butter:Test for Dalda in butter. Adulteration of Khoa: Test for starch in Khoa. Adulteration of Paneer: Presence of starch in paneer.

UNIT - III

Hours 6

Testing Adulteration of Oils: Test for sesame oil in other oils Halphen test for cottonseed oil Hexa bromide test for linseed oil Test for added mineral oil Test for added castor oil Detection of argemone oil in other oils Test for rancidity in oils Kries test for testing quality of oil.

UNIT - IV **Hours 6**

Testing Adulteration of Spices: Extraction of flavour. Coriander powder: Test for starch & horse dung power. Chili powder. Test for oil soluble dyes, powdered bran, saw dust and brick powder. Turmeric Powder: Test for metanil yellow and lead chromate polish. Cloves: Test for exhausted cloves. Curry powder: Test for metallic colours.

UNIT - V **Hours 6**

Testing Adulteration in Seeds: Poppy seeds: Test for Amaranthus seeds. Sajeera: Test for sand, stones and other seeds. Mustard seeds: Visual examination. Pepper: Test for papaya seeds. Saffron: Detection of maize cob tendrils. Cumin seeds; Cinnamon: plant bark.

REFERENCE BOOK

- Rapid detection of food adulterants and contaminants Theory and practice. Shyam Narayan Jha. 2015.

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SECOND YEAR	NON MAJOR ELECTIVE - II	CREDIT : 2
SEMESTER - III	VALUE ADDITION IN FOOD	P25BCNM2B

COURSE OUTCOMES

The objectives of this course are to

- To study the process of and provide a new perspective on value adding, with an emphasis towards 'authenticity' in food.
- To understand the details of food preservation technologies and Value addition in food.
- To understand the anti-nutritional factors.

COURSE SPECIFIC OUTCOMES

- To study the dairy products processing and meat and poultry preservation.
- This course delivers new knowledge and provides future direction to small scale farming Producers and the wider food industry.

UNIT - I **Hours 6**

Food Science and Technology: Methods of food preservation, Use of non-thermal technologies, alternate-thermal technologies, infrared biological technologies -antimicrobial enzymes and bacteriocins in food processing.

UNIT - II **Hours 6**

Fruits and Vegetables: Value addition and storage of fruits and vegetables. Fruit juice, jam, jelly, marmalade, squash, candies, tomato sauce, ketchup, and puree, chips, pickles. Dehydrated fruits and vegetables. Fermented foods and beverages from fruit and vegetables.

UNIT - III **Hours 6**

Food Grains, Spices and Plantation Crops: Anti-nutritional factors in food grains and oilseeds. Value added food grain products like breads, biscuits, cakes, doughnuts, buns, pasta goods, extruded, Instant ready mixtures, puffed foods, confectionary products, breakfast cereals, snack foods, malted food products, legume based food products.

UNIT - IV **Hours 6**

Dairy Products: Reconstituted and flavoured milks. Technology of fermented milks. Milk products processing viz. cream, butter, ghee, cheese, condensed milk, evaporated milk, whole and skimmed milk powder, ice cream, khoa, channa, paneer and similar products. Food Fortification: Fortification of bread, pasta, noodles, biscuits, and breakfast cereals. Micronutrient fortification of snack products. Other special fortified products - salt, sugars, milk and oils. Safety limits.

UNIT - V **Hours 6**

Meat Fish and Poultry Products: Meat and poultry preservation like curing, smoking, freezing, canning and dehydration Value addition and by-products utilizations. Factors influencing keeping quality of meat. Processing and preservation of fish and its products. Preservation canning, smoking and freezing of fresh and sea water fish and its products. Utilization of by-products from fish processing industries. Preservation methods of shell eggs and egg products freezing- pasteurization- desugarization. Technology of egg products viz. egg powder, albumen and flakes.

REFERENCE BOOK

- Potter, Food Science.
- Srivastava and Sanjeev Kumar, Fruit and Vegetable Preservation.
- Fennma Karrel, Principles of Food Science, Vol-I by.
- GirdhariLal, Sidhapa and Tandon. Preservation of Fruits and Vegetables.
- Chakraborty AC. Post-Harvest Technology of cereal pulse and oil seeds.
- Sumati R Mudambi. Food Science.
- Pearson AM and Gillett TA (1996). Processed Meats, CBS Publishers.
- Cole DJA and Lawrie RA (1975). Meat, AVI Pub.
- Stadelman WJ, Olson VM, Shemwell GA and Pasch S (1988). Egg and poultry meat processing, Elliswood Ltd.
- Lawrie R. Developments in Meat Science – I &II, Applied Science Pub. Ltd.
- Stadelman WJ and Cotterill OJ (1973). Egg Science and Technology, AVI Pub.
- Bremner HA (2002). Fish as Food. Vol 1 and 2 CRC Press.
- Jhingram VG (1983). Fish and Fisheries of India, Hindustan Pub Corp.
- George Borgstrom. Fish as Food, Vol. I-IV, Academic Press
- Rogestein & Rogestein. Fish Processing Technology,

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SEMESTER – IV

SECOND YEAR	CORE COURSE - X	CREDIT : 5
SEMESTER - IV	ENDOCRINOLOGY	P25BCCC41

COURSE OUTCOMES

The objectives of this course are to

- Study the nature, function and classification of hormones.
- Acquire knowledge on the structure of Thyroid gland, Parathyroid, Adrenal, Thymus and Pineal gland
- Acquire knowledge on the synthesis of their hormones.
- Understand the gastrointestinal hormones and sex hormones.
- Understand the role of hormones in pregnancy and lactation.

COURSE SPECIFIC OUTCOMES

- Acquire knowledge on the Nature and functions of hormones and learn the
- mechanism of hormone action.
- Learn the structure and functions of Thyroid and Parathyroid gland and familiarize on their hormones.
- Understand the hormones secreted by gastrointestinal tracts and adrenal glands and their action on stress management.
- Able to learn the Structure and functions of the adrenal, pineal and thymus glands.
- Know the feedback control of gonadal hormones and their role in the development
- of male and female gametes.

UNIT – I **Hours 18**

Pituitary, Thyroid and Parathyroid Glands: Nature, function and classification of hormones – Feedback control of hormone secretion – Organisation and functions of neuroendocrine systems- Hypothalamo– hypophyseal interactions- Bioactive peptides. Pituitary gland – Structure and functions, role of hormone secretions - Thyroid gland – Structure, function and biosynthesis of thyroid hormone – Parathyroid –Structure and PTH – Calcitonin – Role of hormones in calcium and phosphate metabolism.

UNIT – II **Hours 18**

Gastrointestinal and Adrenal Hormones: Gastrointestinal hormones - their secretion, control and function – Insulin and glucagons – Adrenal hormones and Stress management – Catecholamines as emergency hormones- their role in the regulation of carbohydrate, protein and lipid metabolisms.

UNIT – III **Hours 18**

Adrenal, Pineal and Thymus Glands: Adrenal gland – Structure and role played its hormones in glucose metabolism – Aldosterone and the rennin- angiotensin system – Pineal gland- structure and its influence on reproduction and pigmentation – Thymus gland – Structure and thymic hormones – their functions in brief

UNIT – IV **Hours 18**

Gonadal Hormones: Steroid hormone biosynthesis in the ovary and testis – Hormonal regulation of ovarian cycles in mammals – Folliculogenesis, ovulation, corpus luteum formation and regression – Hormones in pregnancy and lactation. Gonadal steroid action on spermatogenesis and spermiogenesis – Role of hormones in sex accessory gland growth and functions.

UNIT – V **Hours 18**

Hormone Receptors and Regulation: Steroid hormone receptors, intracellular protein receptors, structural organization of receptor protein, hormone binding domain, antigenic domain and DNA binding domain, organizations of functional elements - hormone response elements, positive and negative transcriptional effects of S.R, receptor activation - upregulation and down regulation, apoptosis - steroid hormone action at cell level, multiple endocrine neoplasia – different types.

Reference Book

- Yadav, Text book of Endocrinology, 2009, Sonali Publications, New Delhi.
- Williams, R H. 1981. Text book of Endocrinology, Ed. 6th W. B. Saunders Company, Philadelphia, London.
- George Griffing, Endocrinology, 2015, Stat Pearls Publishing, USA.
- De Groot. 1979. Endocrinology, Vol. 1-3, Grune and Stratton, New York.
- Mac E. Hadley, Endocrinology. 1996, Prentice Hall
- M.P. Goswami, Endocrinology and Molecular Cell Biology, 2013. Gaurav book centre Pvt Ltd, Delhi
- Stephen Nussey and Saffron Whitehead. 2001. Endocrinology - An Integrated Approach, Oxford: BIOS Scientific Publishers; 2001.
- Bondy P. K. and Rosenberg L.E. 1974. Duncan"s disease of Metabolism – Genetics, Metabolism and Endocrinology. W.B. Saunders Co., Philadelphia, London.

Web - Resources

- https://www.elsevier.com/books/molecular-endocrinology/bolander/978-0-12-111232-5.
- https://www.amazon.in/Molecular-Endocrinology-Franklyn-Bolander-ebook/dp/
- B01D4CI1AQ.
- https://www.amazon.in/Molecular-Endocrinology-Human-Genetics-ebook/dp /B01E3EUF8U.

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SECOND YEAR	CORE COURSE - IX	CREDIT : 5
SEMESTER - IV	PHARMACEUTICAL BIOCHEMISTRY	P25BCCC42

COURSE OUTCOMES

- To understand the different types of bioinformatic tools for drug discovery.
- To get an overview of how different bioinformatic tools aid in the process of target
- identification, drug screening and quantitative structure activity relationship.
- To assimilate the involvement of different metabolic pathways involved in drug metabolism and correlate their involvement in elimination process

COURSE SPECIFIC OUTCOMES

- To understand the biochemical basis of drug action at the target tissue.
- To understand different phases in drug clinical trials and its assessment.

UNIT - I **Hours 18**

Drug discovery and development, drug target identification and validation, hit identification, General principles of screening, correlations between various animal models and human situations, Correlation between in-vitro and in-vivo screens; Special emphasis on cell-based assay, biochemical assay, radiological binding assay, Pharmacological assay, in vitro, in vivo & Ex-vivo experiments, lead optimization, preclinical studies.

UNIT - II **Hours 18**

Bioinformatics approaches for drug development: Identification of potential molecules, chemical compound library preparation, Identification of target in pathogen, Ligand & protein preparation, Molecular docking, Binding free energy estimation, High throughput virtual screening, Docking protocol validation and enrichment analysis, Single point energy calculation, Pharmacokinetics and Pharmacodynamics, ADME & toxicity prediction, Molecular dynamic simulation, Rule of three and five, Lipinsky rule, Pharmacophore development, Quantitative structure activity relationship, 3D-QSAR, Techniques of developing a pharmacophore map covering both ligand based and receptor based approaches.

UNIT - III **Hours 18**

Drug metabolism & interactions: Drug-receptor interactions, receptor theories and drug action, xenobiotics phases (Phase-I, II and III), role of cytochrome P450 oxidases and glutathione S transferases in drug metabolism, factors affecting drug metabolism, Enzymes as a drug target, Kinase inhibitors, ATPase inhibitors, drug protein interaction, Drug DNA interaction. Basic ligand concepts agonist, antagonist, partial agonist, inverse agonist, efficiency and potency. Forces involved in drug receptor complexes. Receptor classification – the four super families. Receptor binding assays- measurement of Kd, Bmax and IC_{50}.

UNIT - IV **Hours 18**

Biochemical mode of action of antibiotics- penicillin and chloramphenicol, actions of alkaloids, antiviral and antimalarial substances. Biochemical mechanism of drug resistance-sulphonamides. Drug potency and drug efficacy. General principles of chemotherapy: chemotherapy of parasitic infections, fungal infections, viral diseases. Introduction to immune modulators and chemotherapy of cancer.

UNIT - V **Hours 18**

Clinical trials (Phase-I, II, III and IV clinical trial). Main features of clinical trials, including methodological and organizational considerations and the principles of trial conduct and reporting. Key designs surrounding design, sample size, delivery and assessment of clinical trials.

REFERENCE BOOK

- Charifson P. Practical Application of Computer-Aided Drug Design, Ed., Marcel Dekker Inc.
- Kubinyi H, Ledien 3D. QSAR in Drug Design: Theory, Methods and Applications, Ed.
- Pharmaceutical Profiling in Drug Discovery for Lead Selection, Borchardt RT, Kerns, EH,
- Lipinski CA, Thakker DR and Wang B (2004). AAPS Press.
- HP Rang (2006). Drug Discovery and Development, Technology in Transition. Elsevier Ltd 1st edition.
- Kenakin TP (2012). Pharmacology in Drug Discovery, Elsevier, 1st Edition.

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SECOND YEAR	SKILL ORIENTED COURSE - I	CREDIT : 2
SEMESTER - IV	MEDICAL LAB TECHNIQUES	P25BCSC1

COURSE OUTCOMES

The main objectives of this course are to

- Impart knowledge on specimen collection and disposal of waste.
- Acquaint knowledge on collection, preservation and transfusion of blood.
- Quantify the biomolecules in biological sample.
- Understand the significance of various tests and their interpretation in diseased conditions.
- Acquaint knowledge on enzymes, hormones and Immunoglobulins as markers for diagnosis.

COURSE SPECIFIC OUTCOMES

- Collect & preserve of biological samples.
- Estimate the various constituents in biological sample.
- Perform the routine procedures adopted in blood bank.
- Analyze and interpret the values for both normal and disease conditions.
- Assay the enzymes and hormones &interpret clinical implications.

UNIT - I

Hours: 18

Collection, transport, analysis of specimen – blood, routine urine, faeces, sputum, semen, CSF Documentation of samples & results. Disposal of laboratory/ hospital waste - Non-infectious waste, biomedical waste, infected sharp waste disposal, infected non sharp disposal – colour coding as per guidelines.

UNIT - II

Hours: 18

Determination of Blood group and Rh factor -Basic blood banking procedures- cross matching, screening test. Blood transfusion and hazards.

UNIT - III

Hours: 18

Estimation of blood sugar – Enzymatic method, HbA1C, Qualitative and quantitative analysis of urine sample- NPN-urea, uric acid, creatinine. Mineral, vitamin and CSF analysis.

UNIT - IV

Hours: 18

Immuno diagnostics -Widal test, VDRL test, ASO, RA, CRP and Complement Fixation Test. RIA, ELISA, Skin test – Montaux and Lepramin test.

UNIT - V **Hours: 18**

Assay of clinically important enzymes- Estimation of clinically important hormones – Insulin, Thyroid and Reproductive hormones and its clinical significance.

REFERENCE BOOKS

- Kanai L Mukherjee and Anuradha Chakravarthy (2022). Medical Laboratory Technology 4th edition,Vol I.
- Ramnik Sood (2006). Text Book of Medical Laboratory Technology, Jaypee Publishers.
- Tietz N (2018). Fundamentals of Clinical Chemistry and Molecular Diagnostics 8th
- edition, WB. Saunders Company.

WEB RESOURCES

- https://www.youtube.com/watch?v=QNYlX5Ne9lQ.
- https://www.slideshare.net/doctorrao/agglutination-tests-and-immunoassys.
- https://microbenotes.com/introduction-to-precipitation-reaction/.

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SUBJECT CODE

E.G.S. PILLAY ARTS & SCIENCE COLLEGE

(An Autonomous Institution, Affiliated to Bharathidasan University, Tiruchirappalli)

Nagore Post, Nagapattinam – 611 002, Tamilnadu.

MSC BIOCHE

Sem	Course	Course	Subject	Course Code
I	Core Course – I (CC)	CC1	Chemistry of Biomolecules	P25BCCC11
	Core Course – II (CC)	CC2	Biochemical Techniques	P25BCCC12
	Core Course – III (CC)	CC3	Physiology & Cell Biology	P25BCCC13
	Core Practical – I (CP)	CP1	Biomolecules & Biochemical Techniques (P)	P25BCCP1
	Elective Course – I(EC)	EC1	(A) Human physiology	P25BCE1A
	Elective Course – I(EC)	EC1	(B) Biostaticstics	P25BCE1B
II	Core Course – IV(CC)	CC4	Enzymology	P25BCCC21
	Core Course – V (CC)	CC5	Molecular Biology	P25BCCC22
	Core Course – VI(CC)	CC6	Metabolism & Regulation	P25BCCC23
	Core Practical – II (CP)	CP2	Enzymology&Molecular Biology(P)	P25BCCP2
	Elective Course – II	EC1	(A)Bioinformatics	P25BCE2A
	Elective Course – II	EC1	(B)Nanotechnology	P25BCE2B
	Non-Major Elective Course I	NME1	(A)Nutritional Biochemisty	P25BCNM1A
	Non-Major Elective Course I	NME1	(B)Herbal Medicne	P25BCNM1B
III	Core Course – VII (CC)	CC7	Immunology	P25BCCC31
	Core Course – VIII (CC)	CC8	Research Methodology	P25BCCC32
	Core Course – IX (CC)	CC9	Clinical Biochemistry	P25BCCC33
	Core Practical – III (CP)	CP3	Clinical Biochemistry & Immunology(P)	P25BCCP3
	Elective Course – III (EC)	EC3	(A) Genomics & proteomics	P25BCE1A
	Elective Course – III (EC)	EC3	(B) Bioethics and IPR	P25BCE1B
	Internship	IS	Internship	P25BCIS
	Non-Major Elective Course II	NME2	(A) Detection methods of food adulteration	P25BCNM2A
	Non-Major Elective Course II	NME2	(B) Value addition in food	P25BCNM2B
IV	Core Course – X (CC)	CC10	Endocrionology	P25BCCC41
	Core Course-XI (CC)	CC11	Pharmaceutical Biochemistry	P25BCCC42
	Skill Oriented Course-I(SOC)	SC	Medical Laboratory Techniques	P25BCSC1
	Project Work	PW	Dissertation = 80 Marks, Viva = 20 Marks	P25BCPW

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